

IEM 4127 - Industrial Management: Previous 6 Years Question Analysis (2018-2023)

Course: IEM 4127 - Industrial Management
Analysis Period: 2018-2023 (6 Years)

Rating System

Rating System:

- ★★★★★ = Appeared 5+ times (**MUST PREPARE**)
 - ★★★★ = Appeared 4 times (HIGH PRIORITY)
 - ★★★ = Appeared 3 times (IMPORTANT)
 - ★★ = Appeared 2 times (MODERATE)
 - ★ = Appeared 1 time (LOW PRIORITY)
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SECTION A: Management Theory & Concepts

1. Management Fundamentals & Functions

Question Type	Years Appeared	Rating
Define management / industrial management	2018, 2019, 2020, 2021, 2022	★★★★★
Management as a dynamic process	2018, 2022, 2023	★★★
Network of management functions	2019, 2021	★★
Management vs Administration	2018, 2021	★★

2. Managerial Skills & Roles

Question Type	Years Appeared	Rating
Managerial skills required for effective	2018, 2019, 2021, 2022,	★★★★★

Question Type	Years Appeared	Rating
leadership	2023	
Managerial roles in organization	2019	★

3. Management Theories & Schools

Question Type	Years Appeared	Rating
Scientific Management Theory (Taylor/Fayol)	2018, 2020, 2022	★ ★ ★
Contingency Theory / Systems Theory School	2018, 2021, 2023	★ ★ ★
Bureaucracy Theory School	2019, 2020	★ ★
Hawthorne experiment / Human relations	2019	★

4. Organizational Structure

Question Type	Years Appeared	Rating
Delegation - objectives, advantages, barriers	2020, 2021, 2022, 2023	★ ★ ★ ★
Line, Line-Staff, Functional organizations	2021, 2022, 2023	★ ★ ★
Decentralization	2019, 2020	★ ★
Committee - functions and effectiveness	2021, 2023	★ ★
Flat vs Tall structure	2021	★

5. Motivation Theories

Question Type	Years Appeared	Rating
Herzberg's Two-Factor Theory (+ Maslow comparison)	2018, 2019, 2021, 2023	★ ★ ★ ★
Group dynamics / Positive motivation	2019, 2020	★ ★
Murray's theory of personality	2018	★

6. Leadership

Question Type	Years Appeared	Rating
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Question Type	Years Appeared	Rating
Leadership styles (Democratic, Authoritarian, Laissez-Faire)	2020, 2022, 2023	★ ★ ★
Manager vs Leader differentiation	2022	★

7. Wage & Incentive Plans

Question Type	Years Appeared	Rating
Bedaux Plan calculation	2019, 2023	★ ★
Halsey Plan calculation	2021, 2022	★ ★
Wages vs Salaries / Job evaluation	2019	★

8. Breakeven Analysis

Question Type	Years Appeared	Rating
Breakeven point definition & calculation	2018, 2019, 2021, 2022, 2023	★ ★ ★ ★ ★
Upper and lower breakeven points	2019, 2021, 2023	★ ★ ★
Profit calculation at given units	2021, 2022, 2023	★ ★ ★

9. Depreciation

Question Type	Years Appeared	Rating
Straight-Line (SL) depreciation	2019, 2021, 2022, 2023	★ ★ ★ ★
Double-Declining Balance (DDB) depreciation	2019, 2021, 2022	★ ★ ★
Book value calculation	2020, 2021, 2023	★ ★ ★

SECTION B: Operations Management

10. Forecasting

Question Type	Years Appeared	Rating
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Question Type	Years Appeared	Rating
Linear trend equation / prediction	2018, 2019, 2020, 2021, 2022	★★★★★
Define forecasting / Quantitative vs Qualitative	2018, 2019, 2021, 2022	★★★★
Exponential smoothing vs Moving average	2018, 2019, 2021	★★★
Forecast accuracy evaluation (MAD, tracking signal)	2021, 2022, 2023	★★★
Steps in forecasting process	2019	★

11. Inventory Management

Question Type	Years Appeared	Rating
EOQ vs EPQ comparison	2018, 2019, 2022, 2023	★★★★
Quantity discounts / Order size optimization	2018, 2019, 2020, 2022	★★★★
EOQ model derivation with assumptions	2018, 2019, 2023	★★★
A-B-C approach to inventory control	2018, 2019, 2021	★★★
EPQ calculation	2021, 2022	★★
Safety stock / Reorder point (ROP)	2022, 2023	★★

12. MRP (Material Requirements Planning)

Question Type	Years Appeared	Rating
MRP calculation with product structure tree	2019, 2021, 2022, 2023	★★★★
Define MRP / Components (BOM, MPS, Inventory records)	2019, 2022, 2023	★★★
Lot-for-lot ordering	2019, 2022, 2023	★★★
Planned order releases/receipts	2021, 2023	★★

13. Scheduling & Sequencing

Question Type	Years Appeared	Rating
Johnson's Rule (two-machine flow shop)	2019, 2023	★★

Question Type	Years Appeared	Rating
Priority rules (FCFS, SPT, CR, S/O)	2018, 2021	★ ★
Gantt charts	2023	★
PERT vs CPM differentiation	2022	★

14. Assignment Problem

Question Type	Years Appeared	Rating
Assignment method (Hungarian method)	2018, 2019, 2020, 2021, 2022	★ ★ ★ ★ ★
Dummy activity definition	2018, 2022	★ ★

15. Project Management / CPM-PERT

Question Type	Years Appeared	Rating
Critical path determination	2021, 2022	★ ★
Slack time calculation	2021, 2022	★ ★
Network diagram construction (AON/AOA)	2018, 2022	★ ★
Probability of project completion (PERT)	2018	★
Project life cycle	2022	★

16. Facility Location

Question Type	Years Appeared	Rating
Brown-Gibson approach / Location analysis	2018, 2022, 2023	★ ★ ★
Objective vs Subjective factors	2018, 2023	★ ★
Reasons for seeking new location	2020	★

17. Layout

Question Type	Years Appeared	Rating
Product layout vs Process layout	2019, 2021	★ ★

Question Type	Years Appeared	Rating
Layout recommendations for scenarios	2018	★

18. Maintenance

Question Type	Years Appeared	Rating
Corrective vs Breakdown vs Preventive maintenance	2018, 2022	★★

19. Aggregate Planning

Question Type	Years Appeared	Rating
Level capacity vs Chase demand strategy	2018, 2021	★★
Demand and capacity options	2018	★

PRIORITY SUMMARY

TOP 10 MUST-PREPARE TOPICS

Rank	Topic	Rating	Section
1	Define Management / Industrial Management	★★★★★	A
2	Managerial Skills	★★★★★	A
3	Breakeven Analysis (calculation + theory)	★★★★★	A
4	Linear Trend Equation / Forecasting	★★★★★	B
5	Assignment Problem (Hungarian Method)	★★★★★	B
6	Delegation	★★★★	A
7	Herzberg's Two-Factor Theory	★★★★	A
8	Depreciation (SL & DDB methods)	★★★★	A
9	EOQ vs EPQ / Quantity Discounts	★★★★	B
10	MRP Calculation with Product Structure	★★★★	B

QUICK REVISION CHECKLIST

Section A - Theory Topics

- ☐ Definition of Management & Industrial Management
- ☐ Managerial Skills (Technical, Human, Conceptual)
- ☐ Management Theories (Scientific, Contingency, Systems)
- ☐ Organizational Structures (Line, Staff, Functional)
- ☐ Delegation & Decentralization
- ☐ Herzberg's Two-Factor Theory vs Maslow's Hierarchy
- ☐ Leadership Styles
- ☐ Breakeven Point Calculation
- ☐ Depreciation Methods (SL, DDB)
- ☐ Incentive Plans (Halsey, Bedaux)

Section B - Numerical Topics

- ☐ Linear Trend Forecasting
 - ☐ Exponential Smoothing
 - ☐ EOQ & EPQ Derivation and Calculation
 - ☐ Quantity Discount Problems
 - ☐ MRP with Product Structure Tree
 - ☐ Assignment Problem (Hungarian Method)
 - ☐ Johnson's Rule (Two-Machine Sequencing)
 - ☐ CPM/PERT Network Analysis
 - ☐ Brown-Gibson Location Method
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EXPECTED QUESTION PATTERN

Based on 6-year analysis:

Section A (Answer 3 out of 4):

- Management definition + Managerial skills
- Management theory + Organizational structure
- Motivation theory + Leadership
- Breakeven analysis + Depreciation + Incentive plan

Section B (Answer 3 out of 4):

- Forecasting (Linear trend / Exponential smoothing)
- Inventory (EOQ/EPQ + Quantity discounts)
- MRP + Scheduling/Sequencing
- Assignment problem + CPM/PERT + Location

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